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# Tumor markers in gynecology

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# Definition

The US National Cancer Institute (NCI) defines tumor markers as substances found in the blood, urine, stool, or tumor tissue that are used for diagnosis, prediction of the response to treatment, and identification of a relapse.

# Introduction

According to the National Institutes of Health (NIH) Definition Working Group, a biomarker is a characteristic objectively measured and evaluated indicator of normal biological processes, pathogenic processes or pharmacological responses to a therapeutic intervention [B01]. This Working Group further separates markers into those that are a marker of the natural history of a disease and correlate longitudinally with known clinical indices (Type 0 markers) and those that capture the effects of a therapeutic intervention in accordance with its mechanism of action (Type 1 markers). In general, validated biomarkers may have utility at all points in the management of cancer, namely as:

1. Predisposition markers: used to identify those at increased risk of developing cancer, e.g. patients with inherited mutations in BRCA1/2 and DNA MMR genes;
2. Screening markers: used in early detection such as prostate-specific antigen (PSA) in the early detection of prostate cancer;
3. Diagnostic markers: used to define the type, stage or grade of a tumor, e.g. immunohistochemical assays of histological samples;
4. Prognostic markers: used to identify the likely disease course and may therefore direct therapy with low risk groups avoiding therapy and high risk groups receiving intensification of therapy;
5. Predictive markers: used to identify those more (positive predictive marker) or less likely (negative predictive marker) to benefit from treatment, e.g. ER, PR, HER-2 expression.

   [B01]

Biomarkers are also useful in drug development where they may be used as endpoints and act as surrogate markers of outcome, thus allowing earlier read out of efficacy to support further development. Pharmacokinetic markers give information about drug levels, distribution and elimination kinetics. They can be used to assess proof of mechanism, namely whether a therapy is hitting its target, and proof of concept, namely whether it is having its desired effect on the tumor biology. This information is vital in interpreting the results of early phase trials. Biomarkers are also useful to gain a mechanistic understanding of the differential clinical responses and help to understand and gain crucial knowledge from negative as well as positive trials. Therapies fail because they cannot be delivered at therapeutic doses, they do not have the desired effect on the proposed target, the hypothesis and proposed mechanism of action is wrong, the wrong population is included or resistance mechanisms are present or under development. Without appropriate markers it is not possible to understand why a trial is negative. Biomarkers can accelerate drug development and prevent effective therapies being discarded.

A good biomarker must have excellent sensitivity and specificity and should be measurable, reproducible, and, if possible, cost-effective. Sensitivity refers to the ability of the biomarker to correctly recognise patients suffering from the disease, and specificity refers to the ability of the biomarker to correctly exclude healthy individuals without the disease. There is often a trade-off between the two and the extent to which this is acceptable will depend on the clinical situation.

Biomarker development requires rigorous investigation and validation. Numerous challenges exist in the development of biomarkers and there are risks associated with their use and development. The risks incorporate wrong diagnosis or treatment and in regards to the development of a therapy or the marker itself through poor trial design. Therefore, all potential biomarkers need to be extensively validated. Hereby challenges exist due to tumor heterogeneity, sensitivity to variations in sample handling, processing and storage and assay validity. In order to improve the development of biomarkers, Cancer Research UK has developed roadmaps for screening, diagnostic, prognostic/predictive, pharmacological and imaging biomarkers. The National Cancer Research Institute (NCRI), NCI, and European Organisation for Research and Treatment of Cancer (EORTC) have also set up a working group to examine the risks and challenges of incorporating biomarkers into clinical trials and develop a risk-assessment framework as well as a list of useful resources to help principal investigators when developing protocols for biomarker-driven trials [B02].

Tumor markers are identified by means of genome, transcriptome, and proteome analyses, and consist of DNA mutations/deletions, increased/decreased mRNA or protein levels, or protein modifications. [B02]

Secreted proteins are particularly useful because they can be detected in the patient's blood. Nevertheless, tumor markers are only of restricted use, given that almost every healthy person has small amounts of tumor markers in their blood, an increase usually only occurs after extensive tumor involvement, while some patients with carcinoma never exhibit increased tumor marker values.

The optimal detection range for patients with carcinoma is consequently in the range of latent growth, far below the clinically detectable tumor size (number of tumor cells: 10^3-10^9). However, the current detection limit of usable tumor markers is 109, close to the clinical detection size (10^12).

In gynecologic oncology, various tumor markers are available nowadays, some of which are also utilized for other carcinomas (table 1).

Table 1: Gynecological carcinomas and existing tumor markers

| Tumor | Tumor marker | Treatment stage | Starting material |
|---|---|---|---|
| Epithelial ovarian carcinoma (non-mucinous) | BRCA1/2 | Prognosis, treatment option (targeted therapy; PARP inhibitors) | Tumor tissue |
| Epithelial ovarian carcinoma (non-mucinous) | HRD | Prognosis, treatment option (targeted therapy; PARP inhibitors) | Tumor tissue |
| Epithelial ovarian carcinoma (non-mucinous) | CA-125 | Diagnosis, response to treatment, relapse | Blood |

| Tumor | Tumor marker | Treatment stage | Starting material |
|---|---|---|---|
| Epithelial ovarian carcinoma (non-mucinous) | Human Epididymis 4 (HE4) | Diagnosis, response to treatment, relapse | Blood |
| Mucinous (epithelial) ovarian carcinoma | CA19-9 | Diagnosis, response to treatment, relapse | Blood |
| Sex cord/stromal tumors (ovary; Sertoli/leydig cell tumors) | Estrogen/ testosterone levels | Diagnosis, relapse | Blood |
| Sex cord/stromal tumors (ovary; granulosa cell tumors) | Inhibin | Diagnosis, relapse | Blood |
| Germ cell tumors (ovary; dysgerminoma) | Lactate dehydrogenase (LDH) | Diagnostics (stage), prognosis, response to treatment | Blood |
| Germ cell tumors (ovary; non-dysgerminoma) | Alpha-fetoprotein (AFP) | Diagnostics (stage), prognosis, response to treatment | Blood |
| Choriocarcinomas, germ cell tumors (ovary; non-dysgerminoma) | Beta-human chorionic gonadotropin (Beta-hCG) | Diagnostics (stage), prognosis, response to treatment | Blood, urine |
| Endometrial carcinoma | POLE, MMR, p53 | Prognosis, response to treatment | Tumor tissue |
| Cervical carcinoma | HPV | Prognosis, response to treatment | Tumor tissue |
| Cervical carcinoma | PD-L1 | Treatment option (targeted therapy; Pembrolizumab) | Tumor tissue |
| Breast carcinomas | CA15-3 | Treatment outcome, relapse | Blood |

| Tumor | Tumor marker | Treatment stage | Starting material |
|---|---|---|---|
| Breast carcinomas | Estrogen/ progesterone receptor (ER/PR) | Treatment option (targeted therapy; tamoxifen, aromatase inhibitors) | Tumor tissue |
| Breast carcinomas | HER2/new | Treatment option (targeted therapy; trastuzumab) | Tumor tissue |
| Breast carcinomas | HRD | Prognosis, response to treatment | Tumor tissue |
| Breast carcinomas | Oncotype | Prognosis, response to treatment | Tumor tissue |
| Vulvar carcinoma | HPV | Prognosis, response to treatment | Tumor tissue |
| Vulvar melanoma | c-KIT | Diagnostic, assess treatment options (targeted therapy; Iressa) | Tumor tissue |

# Ovarian carcinoma tumor markers

1. Testing for BRCA1/2 mutations and/or genomic instability (HRD) is recommended for all patients diagnosed with epithelial ovarian carcinoma.
2. Approx. 50 % of epithelial ovarian carcinomas are HRD+ and thus benefit from treatment with PARP inhibitors (PARPis).

Globally, ovarian carcinoma accounts for approx. 4 % of cancer incidence and mortality in women, with 313’959 new cases diagnosed and 200’000 deaths each year [B03]. We will mainly focus on validated tumor markers recommended for clinical use in cases of epithelial ovarian carcinoma, which accounts for more than 90 % of all malignant ovarian carcinoma diagnosis. [B03]

## CA-125

CA-125 (cancer antigen 125) is the only FDA-approved tumor marker for the identification of epithelial ovarian carcinoma. CA-125 is a transmembrane glycoprotein. Approx. 80 % of patients with epithelial ovarian carcinoma have elevated CA-125 levels of over 35 U/mL [B04, B05]. [B04][B05]

A notable NCI screening study examined this question for various cancers, including ovarian carcinoma [P01, P02]. This major, population-based, randomized study carried out by the NCI investigated whether screening affects tumor patient survival [B02, B03].

A total of 155,000 people aged 55 to 74 years with no history of carcinoma were included at ten screening centers across the United States. Subjects were assigned at random to an intervention arm (screening program) or a control arm (usual care) (inclusion phase 1993-2001: 78’000 women and 76’000 men). The results of the study with regard to ovarian carcinoma detection are sobering: Screening with CA-125 and transvaginal ultrasound increased the percentage of diagnosed ovarian carcinoma, but did not reduce mortality. [B02][B03][P01][P02]

Three more major population studies have examined the use of population screening with regard to ovarian carcinoma: one American, one Japanese and one British.

- The University of Kentucky study of 25’327 women investigated transvaginal ultrasound (TVUS) used alone and found a higher rate of survival in the screening arm ($p < 0{,}001$; 74,8 % vs. 53,7 %) [P03].
- A Japanese randomized controlled trial showed an increase in the proportion of

early ovarian carcinoma (FIGO I) through screening by means of gynecological examination, TVUS and CA-125 testing, with good sensitivity (77,1 %).
- The largest randomized controlled trial with two screening arms is the United Kingdom Collaborative Trial of Ovarian Cancer Screening (UKCTOCS). The study included 98’305 women and investigated screening with ultrasound alone versus CA-125 testing followed by ultrasound testing (with subsequent interpretation of both results using the Risk of Ovarian Cancer Algorithm) [P04]. [P03][P04]

The results showed more promising sensitivity (89,4 %) for the combined screening arm, but as with the Japanese study, mortality figures were no better for cases detected earlier [P05] (table 2).

The UKCTOCS is particularly interesting as the tumor marker CA-125 was tested on a patient-specific basis, i.e. determining an individual baseline per patient and a deviation of at least doubling as the cutoff [P05]. In contrast, the Japanese study and the PLCO study always used 35 U/mL as the cutoff for CA-125. A retrospective analysis of the PLCO study addressed this question and investigated whether longitudinal CA-125 determination and deviation from baseline improved detection vs. cutoff. This showed that this method allowed ovarian carcinoma to be identified on average 10 months earlier and that CA-125 levels on diagnosis were under the cutoff of 35 U/mL for 42 % of women (median: 20 U/mL) [P06]. [P05][P06]

Currently, it is clear that:

- Annual CA-125 determination and TVUS is ineffective in the general and high-risk populations and should not be carried out.
- For other gynecological carcinomas, no data on screening with CA-125 is available and it is therefore clearly not indicated.

Given the lack of data, gynecologists should therefore suggest the following method to their patients:

1. Identification of at-risk patients with hereditary risk of carcinoma
2. Consider patient symptoms
3. Take into consideration individual deviations from baseline with regard to CA-125 levels
4. Do not use a cutoff (35 U/mL).

CA-125 is often used in combination with a complete clinical examination and imaging to identify ovarian carcinoma. A useful tool in case of suspected epithelial ovarian carcinoma is the Risk of Malignancy Index, which is based on the multiplication of the patient's menopausal status (1-premenopausal, 3-postmenopausal), transvaginal ultrasound findings (0-no abnormalities, 1-one abnormality, 2-two abnormalities, 3-three abnormalities) and the absolute CA-125 value [B06]. [B06]

In a gynecological carcinoma center with a preselected patient population, this gave [P07]:

- sensitivity of 84 %
- specificity of 77 %
- a positive predictive value of 76 %
- a negative predictive value of 85 %
  [P07]

In our own retrospective study of 5'218 patients who received ultrasound for adnexal tumors at a Swiss university hospital during a five-year period (1998-2012), we also obtained a sensitivity of 85 %, but a specificity of 92 % and a negative predictive value of 98 % for a non-preselected patient population [P08]. [P08]

Caution should be exercised with regard to newly commercialized tumor markers that only show minimal benefit over existing markers. The ovarian marker HE4 is a second marker with an identical detection profile to CA-125, but with a lower level of dispersion, meaning that the detection of benign conditions (e.g. endometriosis) is reduced [B07, B08]. [B07][B08]

In a Swiss cohort of 169 patients, we have been able to show that the expression patterns of HE4 and CA-125 are identical, with the minor exception of HE4 having a lower level of dispersion, resulting in less detection of benign diseases (e.g. endometriosis) [P09]. [P09]

Use of HE4 in addition to CA-125 should only be used in young patients with elevated CA-125 and suspected endometriosis, as reduced HE4 can confirm this diagnosis. CA-125 testing is also a useful marker for response to chemotherapy and is therefore determined as a baseline value alone or in combination with computed tomography after completion of treatment [B09]. Subsequently, repeat testing should only be carried out in case of symptoms [B10]. [B09][B10]

An American study has shown that patients for whom 3-monthly CA-125 testing was carried out showed no difference in survival versus control patients for whom tumor markers were only measured in case of symptoms. Rather, the results of this study showed that the only effect of 3-monthly tumor marker testing was that patients received significantly more chemotherapy cycles [P10].

After completion of treatment, the baseline value for an individual patient should therefore be determined and then repeat testing should only be carried out in case of symptoms [P10]. [P10]

## BRCA 1/2

There is a hereditary risk of carcinoma in carriers of BRCA1 and BRCA2 (BRCA1/2) gene mutations.

- In families with BRCA1/2 mutations, there is a particular focus on paternal/ maternal hereditary breast, ovarian, prostate and pancreatic cancers [P11, P12, P13].
- Triple negative breast cancer and poorly-differentiated high-grade serous ovarian carcinomas are BRCA1/2 mutation-associated carcinomas [P14, P15].
- As it is now known that this ovarian carcinoma originates in the fimbriae, bilateral risk-reducing salpingo-oophorectomy (RRSO) is the only method of prevention and should therefore be carried out after childbearing is complete as this reduces the risk of ovarian carcinoma by 98 % and fallopian tube cancer by 50 % [P16]. For women with a BRCA1 mutation, RRSO is typically recommended between the ages of 35 and 40, or after childbearing is complete. For women with a BRCA2 mutation, the recommended age for RRSO is slightly later, generally between 40 and 45, or after childbearing is complete.
  [P11][P12][P13][P14][P15][B39]

Germline BRCA1/2 mutations are found in approx. 17 % of high-grade ovarian carcinomas, most often with serous ovarian carcinoma [B11, B12]. Tumors with a BRCA1/2 mutation show impaired DNA damage repair. Homologous recombination (HR) is a pathway for DNA double-strand break repair [B13] (figure 1). The repair of strand breaks by HR contributes to genomic stability and thus plays a role in preventing malignant changes. Inactivation of the HR pathway is most often caused by genetic or epigenetic changes to the BRCA1/2 genes, but can also be a result of changes to other genes that code for proteins involved in HR DNA repair. BRCA1/2 gene mutations are the best-characterized prognostic tumor markers for epithelial ovarian carcinoma: several studies have shown that patients with BRCA1/2 mutations respond better to platinum-based chemotherapy and have better rates of survival [B14, B15, B16, B17]. Furthermore, PARPis show greatest effect on patients with BRCA1/2 mutations. Screening for BRCA1/2 mutations is recommended for all patients diagnosed with epithelial ovarian carcinoma [B18, B19, B20, B21, B22, B23]. Patients with high-grade serous ovarian carcinoma and a germline or somatic BRCA1/2 mutation should be treated with a PARPi after response to platinum-based chemotherapy [B09]. [B09][B11][B12][B13][B14][B15][B16][B17][B18][B19][B20][B21][B22][B23]

## HRD

Homologous recombination deficiency (HRD) status, usually evaluated by measuring the extent of DNA scarring in the tumor genome, is another prognostic marker for epithelial ovarian carcinoma. At diagnosis, ~50 % of epithelial ovarian carcinomas present defective HR DNA repair as a result of genetic or epigenetic changes to genes in the HR pathway, and therefore respond well to chemotherapy and PARPis and have longer survival [B20, B21, B22, B23, B24].
Genomic instability (HRD) screening is recommended [B09]. This identifies a subgroup of patients who are BRCA1/2 wild type but could still benefit from PARPi treatment. Patients with a positive HRD test and partial or complete response to platinum-based first-line treatment with or without bevacizumab should receive maintenance therapy with a PARPi [B09]. [B09][B20][B21][B22][B23][B24]

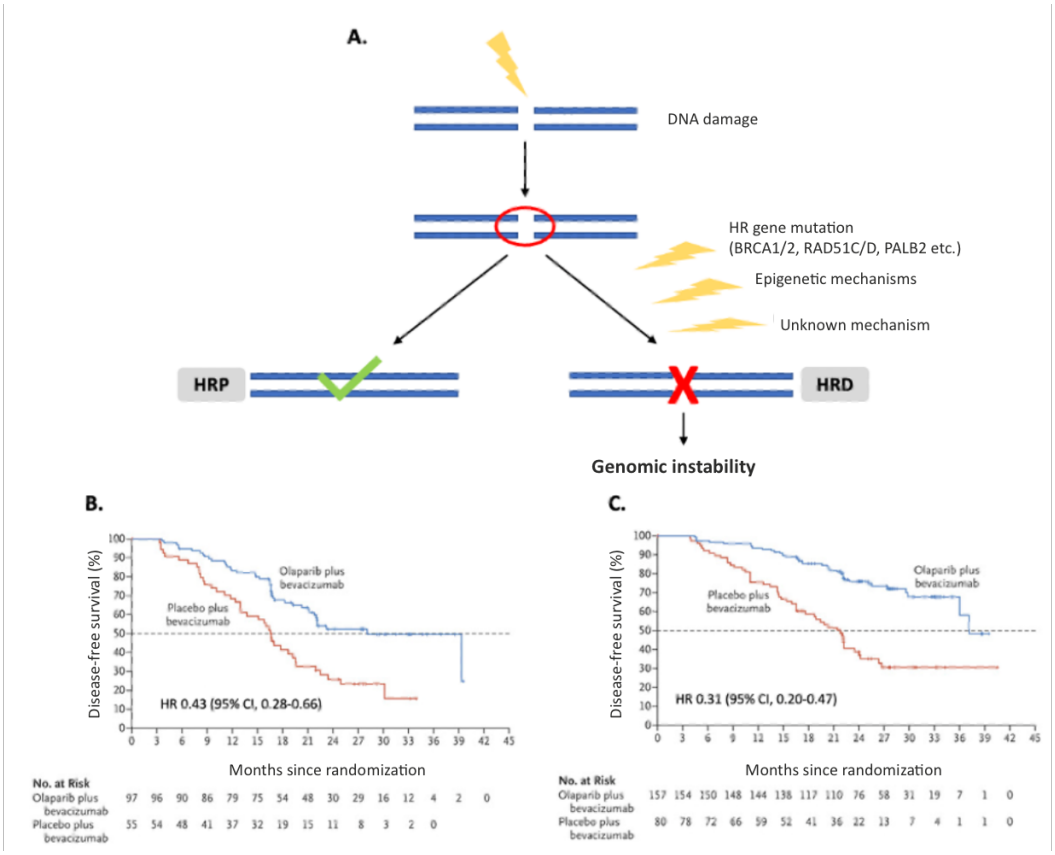


Figure 1: A – Description of homologous recombination deficiency and Kaplan-Meier survival curves comparing Olaparib plus Bevacizumab vs. Placebo plus Bevacizumab as first-line maintenance therapy in patients with high-grade ovarian carcinoma with B – homologous recombination deficiency and C – BRCA mutation.
Legend: HR – Homologous Recombination, HRD – Homologous Recombination Deficiency, HRP – Homologous Recombination Proficiency, HR – Hazard Ratio, CI – Confidence Interval, No. at Risk –Number at Risk (B and C adapted from [B20]). [B20]

# Endometrial carcinoma tumor markers

The molecular classification of endometrial carcinomas helps establish the relative risk of recurrence and plays a role in deciding on adjuvant therapy.

In recent years, it has become increasingly clear that the traditional classification of endometrial carcinomas leads to heterogeneous molecular groups and prevents both the progression and the implementation of precision medicine [B26]. However, there is a simple molecular endometrial carcinoma classification that shows good prognostic value for survival and response, and has recently been included in the European Society of Medical Oncology (ESMO) guidelines to facilitate decision-making on adjuvant treatment for endometrial carcinoma [B27]. [B26][B27]

## POLE, MMR and p53

The Cancer Genome Atlas (TCGA) described four molecular endometrial carcinoma subgroups based on a combination of somatic mutation load and copy number alterations [B28]. These four molecular groups are 1) POLE-mutated with a high mutation load (>100 mutations/megabase (mut/Mb)) with pathogenic variations in the exonuclease domain of DNA polymerase epsilon (POLE), 2) dMMR with moderate mutation load (10-100 mut/Mb), mismatch repair deficient with microsatellite instability, 3) pMMR p53-abnormal with high somatic copy-number, MMR-proficient with pathogenic TP53 variants and 4) pMMR p53-wild type with low somatic copy-number, MMR-proficient with no TP53 variants and no microsatellite instability. (Figure 2) The integration of this molecular classification with established clinical-pathological factors has led to an updated risk classification system in order to establish the relative risk of recurrence [B27]. POLE-mutated endometrial carcinoma shows the best prognosis, and p53-abnormal the worst prognosis and the worst response to therapy regardless of risk group, type of adjuvant treatment, tumor type and grade [B28]. As a result, patients with stage I-II POLE-mutated tumors are now classified as low-risk of recurrence, unlikely to benefit from adjuvant treatment [B29, B30, B31]. Furthermore, it has been shown that p53-abnormal endometrial carcinoma derives the greatest benefit from additional chemotherapy as adjuvant treatment [B29].
It has been shown that molecular classification with identification of equivalent subgroups can also be achieved with the established immunohistochemical (IHC) markers (MLH1, MSH2, MSH6, PMS2 and p53) in combination with targeted tumor sequencing (POLE hotspot analysis) [B29, B30, B31, B32, B33, B34, B35]. Comprehensive sequencing can therefore be avoided and classification also automatically serves as pre-screening for Lynch syndrome. This should be carried out for all endometrial carcinomas regardless of histological type [B27]. [B27][B28][B29][B30][B31][B32][B33][B34][B35]

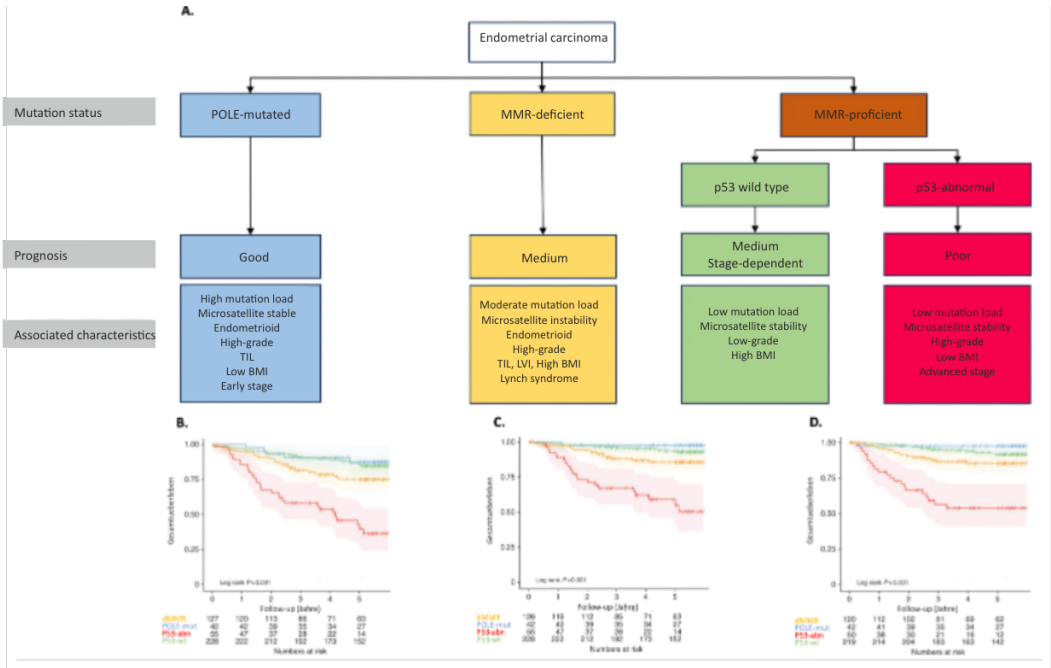


Figure 2: A – Overview of molecular classification of endometrial carcinoma and B – Kaplan-Meier survival curve after molecular classification.
Legend: POLE – Polymerase Epsilon, MMR – Mismatch Repair, TIL – Tumor Infiltrating Lymphocytes, BMI – Body Mass Index, LVI – Lymphovascular Invasion, dMMR – deficient Mismatch Repair, POLE-mut – POLE-mutated, P53-mut – P53-mutated, P53-wt – P53 wild type (adapted from [B27, B30]).

# Cervical carcinoma tumor markers

1. HPV is a prognostic factor for tumor subtype, disease-free survival and overall survival and should be taken into account when deciding on fertility-preserving treatments for cervical carcinoma.
2. PD-L1 is not a good prognostic tumor marker for response to treatment with immune checkpoint inhibitors in cases of cervical carcinoma.

Cervical carcinoma is the fourth most commonly diagnosed cancer and the fourth most common cause of cancer death in women, with the overwhelming majority of deaths occurring in low- and middle-income countries [B03]. The major geographical differences in cervical carcinoma reflect differences in the availability of screening and the prevalence of human papillomavirus (HPV) infection. Squamous cell carcinoma is the most common subtype (70 to 80 %), followed by adenocarcinoma (20 to 25 %) and other epithelial tumors [B36]. Early-stage cervical carcinoma often has no symptoms and is most often detected on cervical screening. Treatment then depends on tumor size, location and stage and the following tumor markers. [B03] [B36]

## HPV

Cervical carcinoma forms in the cervical tissue and is most often caused by HPV infection, with HPV 16/18 responsible for at least two-thirds of cervical carcinomas. Thanks to highly effective primary (HPV vaccination) and secondary (screening) preventive measures, cervical carcinoma is almost completely preventable [B03]. HPV vaccination leads to an annual reduction of 90 % in cervical carcinoma cases and more than 85 % of vaginal, vulvar, anal and penile cancers [B36]. Abnormal cervical cytology or a positive HPV test should lead to colposcopy and biopsy or excision procedures such as loop electrosurgical excision (LEEP) or conization.
For all invasive cervical carcinomas in situ, additional testing is required to confirm HPV involvement. The most widespread technique is p16 immunohistochemistry. Alternatively, HPV E6/E7 gene DNA or mRNA can be identified using in situ hybridization or PCR-based techniques. This shows ~10 % to be HPV-negative. Unlike HPV-associated cervical carcinoma, this is often diagnosed at an advanced stage, not usually squamous cell carcinoma and associated with poorer disease-free survival and overall survival [B37, B38, B39].
This has also led to the modification of ESGO/ESTRO/ESP recommendations for fertility-preserving therapy for cervical carcinoma. Fertility-preserving therapy with preservation of the ovaries is an oncologically-acceptable alternative to radical hysterectomy for young patients with cervical carcinoma < 2 cm who wish to retain the ability to have children. This should be discussed with patients with squamous cell carcinoma and can also be considered in cases of HPV-associated adenocarcinoma, but is not recommended for HPV-negative adenocarcinoma [B40]. Before fertility-preserving therapy is started, an oncofertility consultation and discussion as part of a multidisciplinary tumor board is recommended. After fertility-preserving therapy, follow-up care should include regular HPV tests (after 6-12 and 24 months). Circulating cell-free HPV DNA in plasma is also currently being investigated as a new tool in evaluating response to treatment and monitoring cervical carcinoma [B41]. [B03][B36][B37][B38][B39][B40][B41]

## PD-L1

Targeted therapies like immune checkpoint inhibitors are a recent development in the treatment of cervical carcinoma. As a result, there is a growing need for predictive assays to achieve reliable patient selection for the therapy with an optimal cost-benefit ratio. Roughly 35 % of squamous cell epithelial carcinomas and 17 % of cervical adenocarcinomas express PD-L1 [B42]. PD-L1 tests are carried out on tumor tissue (either biopsies or surgical samples) to select an immune checkpoint therapy. The phase III study MK-3475-826/KEYNOTE-826 showed the benefit of adding pembrolizumab to chemotherapy with or without bevacizumab for patients with persistent, recurrent, or metastatic cervical carcinoma. The addition of pembrolizumab resulted in significantly longer progression-free and overall survival in patients with a PD-L1 combined positive score (CPS) ≥1, in the intention-to-treat population and in patients with a PD-L1 CPS ≥ 10 [B43]. However, the PD-L1-CPS<1 subgroup did not show an improvement in survival among the PD-L1-selected subpopulations. The addition of the checkpoint inhibitor pembrolizumab to platinum-based chemotherapy ± bevacizumab is currently recommended in patients with PD-L1-positive tumors with a CPS ≥ 1.
In addition, patients who had disease progression after first-line platinum-containing chemotherapy but had not received previous immunotherapy were treated with the PD-1 inhibitor cemiplimab, regardless of their PD-L1 tumor status. Survival was significantly longer with cemiplimab than with single-agent chemotherapy among patients with recurrent cervical cancer after first-line platinum-containing chemotherapy, regardless of PD-L1 status [B44]. [B42][B43][B44]

Figure 3: Kaplan-Meier survival curves comparing Pembrolizumab plus vs. Placebo following platinum-containing chemotherapy ± bevacizumab in patients with persistent, recurrent, or metastatic cervical carcinoma A – regardless of PD-L1 status B – PD-L1 CPS ≥1 and C – PD-L1 CPS ≥10. Legend: CPS – Combined Positive Score, HR – Hazard Ratio, CI – Confidence Interval, No. At Risk – Number at Risk (adapted from [B43]).

Figure 3: Kaplan-Meier survival curves comparing Pembrolizumab plus vs. Placebo following platinum-containing chemotherapy ± bevacizumab in patients with persistent, recurrent, or metastatic cervical carcinoma A – regardless of PD-L1 status B – PD-L1 CPS ≥1 and C – PD-L1 CPS ≥10.
Legend: CPS – Combined Positive Score, HR – Hazard Ratio, CI – Confidence Interval, No. At Risk – Number at Risk (adapted from [B43]).

In summary, immune checkpoint inhibitors show good efficacy in cervical carcinoma regardless of PD-L1 status. It is therefore essential that, in the future, a better tumor marker for identifying a subpopulation of cervical carcinoma patients who will most benefit from treatment with immune checkpoint inhibitors is found.

# Conclusion

When planning treatment for gynecological tumors, comprehensive and precise knowledge of prognostic and predictive factors is essential.

It has been proven that in the age of precision oncology, for certain gynecological tumors we can: 1) identify and treat patients before they develop tumors; 2) predict survival and response to treatment with molecular classification; 3) discover new targeted treatment opportunities; 4) identify patients who will most benefit from specific treatments and 5) improve patient management by reducing over- and undertreatment. Nevertheless, further research is required to identify molecular tumor markers with high specificity and sensitivity that could lead to improvements in personalized treatment and the early identification of gynecological tumors. Molecular classification and stratification according to tumor markers should therefore be encouraged when future studies are being planned, as this will enable the comparison of patient groups with similar/analogous characteristics.

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